Practice workbook · Getting a bracket ready for the binomial series: a(1 + bx)^n

Rewriting a bracket such as (4 + 3x)^(1/2) or (2 - x)^(-3) as a constant times (1 + bx)^n, ready for a binomial expansion.

9 question parts: 1 worked-example retry and 8 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-LOG-02

1. Question 5 · part a · 2 marks

2022 JUN · Paper 01

Worked-example retry · Attempt part a only.

5 Given that $(2 + 3x)^{-1}$ can be expressed in the form $p(1 + qx)^{-1}$ where p and q are constants,

(a) find the value of p and the value of q

(2)

(b) Hence expand $(2 + 3x)^{-1}$ in ascending powers of x up to and including the term in x^3, expressing each coefficient as an exact fraction in its lowest terms.

(3)

$$f(x) = \frac{1 + x}{2 + 3x}$$

(c) Obtain a series expansion for $f(x)$, in ascending powers of x up to and including the term in x^3, expressing each coefficient as an exact fraction in its lowest terms.

(2)

(d) Hence use algebraic integration to obtain an estimate, to 4 decimal places, of

$$\int_0^{0.5} f(x)\,dx$$

(4)

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1. Question 5 · part a · 2 marks · Working and answer

Working and answer

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2. Question 6 · part a · 2 marks

2023 JAN · Paper 01

Further practice · Attempt part a only.

6 Given that $(8 + 3x)^{\frac{1}{3}}$ can be expressed in the form $p(1 + qx)^{\frac{1}{3}}$ where p and q are constants,

(a) find the value of p and the value of q

(2)

(b) Hence, expand $(8 + 3x)^{\frac{1}{3}}$ in ascending powers of x up to and including the term in x^2, expressing each coefficient as an exact fraction in its lowest terms.

(3)

Using the expansion found in part (b) with a suitable value of x

(c) show that $\sqrt[3]{9} \approx \frac{599}{288}$

(2)

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2. Question 6 · part a · 2 marks · Working and answer

Working and answer

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3. Question 6 · part a · 2 marks

2025 JUN · Paper 01

Further practice · Attempt part a only.

6 (a) Show that $\sqrt{4-x} = A\left(1-\frac{x}{B}\right)^{\frac{1}{2}}$ where A and B are integers to be found.

(2)

(b) Hence expand $\sqrt{4-x}$ in ascending powers of x up to and including the term in x^3
Give each coefficient as an exact fraction in its lowest terms.

(3)

(c) Use your expansion with a suitable value of x to obtain an estimate of $\frac{\sqrt{305}}{10}$

Give your answer correct to 5 decimal places.

(3)

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3. Question 6 · part a · 2 marks · Working and answer

Working and answer

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4. Question 9 · part a · 2 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part a only.

9 Given that $\frac{1}{(2-x)^3}$ can be written as $p(1-qx)^{-3}$

(a) find the value of p and the value of q.

(2)

(b) Expand $\frac{1}{(2-x)^3}$ in ascending powers of x up to and including the term in x^3 and express each coefficient as an exact fraction in its lowest terms.

(3)

$$f(x) = \frac{a+bx}{(2-x)^3} \text{ where } a \text{ and } b \text{ are integers}$$

The first three terms of the expansion of $f(x)$ are $\frac{3}{8} - \frac{43}{16}x + cx^2$

(c) Find the value of a and the value of b.

(3)

(d) Find the exact value of c.

(2)

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4. Question 9 · part a · 2 marks · Working and answer

Working and answer

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5. Question 9 · part a · 2 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part a only.

9 (a) Write $\frac{3}{(3-x)^3}$ in the form $a(1-bx)^{-3}$
where a and b are fractions in their lowest terms.

(2)

(b) Expand $\frac{3}{(3-x)^3}$ in ascending powers of x up to and including the term in x^3
Express each coefficient as a fraction in its lowest terms.

(3)

(c) (i) Use a suitable value of x with your expansion in part (b), to obtain an approximation for $\frac{24}{125}$ to 5 decimal places.

(ii) Find the percentage error, to 2 decimal places, of your approximation from the actual value.

(4)

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5. Question 9 · part a · 2 marks · Working and answer

Working and answer

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6. Question 6 · part a · 2 marks

2024 NOV · Paper 01

Further practice · Attempt part a only.

6 (a) Show that

$$\frac{a}{\sqrt{4+bx}} = \frac{a}{2}\left(1+\frac{bx}{4}\right)^{-\frac{1}{2}}$$ where a and b are positive integers.

(2)

The expansion of $\frac{a}{\sqrt{4+bx}}$ in ascending powers of x can be written as

$$P+Qx+Rx^2+Sx^3$$

where P, Q, R and S are rational numbers.

(b) Show that $Q=-\frac{ab}{16}$ and $S=-\frac{5ab^3}{2048}$

and find P and R in terms of a and b, as fractions in their lowest terms.

(4)

Given that $Q=\frac{128}{5}S$ and $R=\frac{9}{256}$

(c) show that $a=3$ and $b=1$

(3)

(d) Hence, using an appropriate value of x, find, to 3 decimal places, an approximate value for $\frac{\sqrt{6}}{2}$

(3)

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6. Question 6 · part a · 2 marks · Working and answer

Working and answer

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7. Question 6 · part a · 2 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part a only.

6 Given that $\sqrt{9-x}$ can be expressed in the form $p(1+qx)^{\frac{1}{2}}$ where p and q are constants

(a) find the value of p and the value of q.

(2)

(b) Hence expand $\sqrt{9-x}$ in ascending powers of x up to and including the term in x^3 expressing each coefficient as an exact fraction in its lowest terms.

(3)

Using the expansion you found in part (b) with a suitable value of x,

(c) find an estimate to 5 decimal places for the value of $\sqrt{\frac{31}{4}}$

(3)

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7. Question 6 · part a · 2 marks · Working and answer

Working and answer

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8. Question 10 · part a · 2 marks

2021 NOV · Paper 02

Further practice · Attempt part a only.

10 (a) Show that $\frac{3}{\sqrt{9-3x}} = \left(1-\frac{x}{3}\right)^{-2}$

(2)

(b) Hence expand $\frac{3}{\sqrt{9-3x}}$ in ascending powers of x up to and including the term in x^3 expressing each coefficient as an exact fraction in its lowest terms.

(3)

$$f(x) = \frac{1+2x}{\sqrt{9-3x}}$$

(c) Find the expansion of $3f(x)$ in ascending powers of x up to and including the term in x^3 expressing each coefficient as an exact fraction in its lowest terms.

(4)

(d) Hence, using algebraic integration, obtain an approximation to 6 significant figures for

$$\int_{0.1}^{0.2} \frac{1+2x}{\sqrt{9-3x}}\,dx$$

(4)

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8. Question 10 · part a · 2 marks · Working and answer

Working and answer

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9. Question 7 · part c · 2 marks

2023 JUN · Paper 02

Further practice · Attempt part c only.

7 (a) Expand $\left(1+\frac{x}{3}\right)^{-3}$ in ascending powers of x up to and including the term in x^3

Where appropriate express each coefficient as an exact fraction in its lowest terms.

(3)

(b) Write down the range of values of x for which your expression is valid.

(1)

(c) Express $(3+x)^{-3}$ in the form $P(1+Qx)^{-3}$ where P and Q are rational numbers whose values should be stated.

(2)

$$f(x)=\frac{(1+4x)}{(3+x)^3}$$

(d) Obtain a series expansion for $f(x)$ in ascending powers of x up to and including the term in x^2

(2)

(e) Hence, using algebraic integration, obtain an estimate of $\int_0^{0.2} f(x)\,dx$

Give your answer to 5 significant figures.

(3)

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9. Question 7 · part c · 2 marks · Working and answer

Working and answer

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